

Classification of Vegetables Using Deep Learning

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Abstract

In this work, we study and present an automated approach for classifying four types of vegetables namely, tomatoes, cucumbers, carrots, and cabbages via deep learning models. We evaluate the performance of two well-known convolutional neural networks (CNNs), ResNet and AlexNet, in accurately identifying and categorizing these vegetables based on their visual features.

1 Introduction

The importance of image classification automation cannot be overemphasized. In agriculture, sorting agricultural produce using the traditional approach could be labor-intensive and time consuming, which, in addition, is prone to human error. In this work, we focus on classifying four important vegetables which includes *tomatoes*, *cucumbers*, *carrots*, and *cabbages* using state-of-the-art convolutional neural networks (CNNs), namely ResNet and AlexNet.

Recent years have witnessed significant progress in deep learning for image classification, with CNNs demonstrating remarkable performance in agricultural applications. However, challenges persist in distinguishing vegetables with similar colors, or textures, especially under varying lighting conditions and orientations. While prior works have explored crop disease detection or fruit classification, limited attention has been given to fine-grained vegetable categorization for post-harvest processing.

2 Related Work

Recent advances in deep learning have revolutionized image classification tasks, including agricultural and food-related applications. This section reviews prior work in vegetable/fruit classification, CNN architectures, and deployment tools like Gradio.

Several studies have leveraged CNNs for crop and food recognition. [1] used a modified VGG16 to classify 15 fruits and vegetables with 94% accuracy, while [2] employed YOLOv4 for real-time tomato detection in smart greenhouses. However, these works focused on singular crops or required complex setups. In this work, we target a diverse set of vegetables

(tomatoes, cucumbers, carrots, cabbages) under varying conditions, mimicking real-world retail or post-harvest scenarios. ResNet and AlexNet have been widely adopted in agricultural domains. [3] demonstrated ResNet50's superiority over AlexNet in leaf disease detection, attributing its success to residual connections mitigating gradient vanishing. Similarly, [4] compared both architectures for fruit quality grading, noting ResNet's higher computational cost. Our study extends these insights to vegetable classification, providing empirical comparisons specific to shape and texture variations.

3 Methodology

3.1 Dataset

This project utilizes the vegetable classification dataset, which is publicly available on Kaggle [5], for training the model. The original dataset comprises 15 vegetable classes. However, for the purpose of this project, we selected four classes: **cabbage**, **cucumber**, **carrot**, and **tomato**.

The dataset is organized into three folders: **train**, **test**, and **validation**.

- In the **train** folder, each of the four classes contains **1,000 images**,
- In the **test** folder, each class contains **200 images**,
- In the **validation** folder, each class contains **200 images**,

This results in a total of **4,800 images**.

To simplify the vegetable classification task and enhance the training process, each image in the dataset has been resized to a dimension of 224×224 pixels, ensuring compatibility with the *ResNet18* architecture used in this study. The images are colored JPG files, and the dataset contains well-balanced samples across the selected classes. It is therefore suitable for training and evaluating deep learning models.

Model training and evaluation were tracked with Weights and Biases (wandb) for real-time metric visualization. The trained model was deployed via a Gradio web interface hosted on Hugging Face Spaces, enabling real-time user interaction. An example of the selected vegetable images used in this study is shown in Figure 1.

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Figure 1: Examples of images from the selected classes: Cabbage, cucumber, carrot, and tomato

3.2 Preprocessing

We organized all the images into folders on Google Drive, mounted the drive in Google Colab, and validated the directory structure. The folders — **train**, **validation**, and **test** — were loaded to facilitate easy data transformation. All images were standardized to a resolution of 224×224 pixels to meet the requirements of the *ResNet18* model. To enhance the model’s generalization ability and improve accuracy, several data transformation techniques were applied to the training set. These included resized cropping, horizontal flipping, and normalization.

3.3 Model Architecture

The *ResNet18* model was implemented in this project because of its efficiency and lightweight architecture with 18 layers, making it faster to train. It serves as a reliable model for image classification in both research and industrial applications, offering good accuracy compared to the *AlexNet* model. For this purpose, PyTorch libraries such as *torchvision* were used to implement ResNet18, contributing to the successful outcome of the project.

4 Results

The model was trained for 10 epochs, showing excellent improvement in training metrics and maintaining 90% accuracy, indicating strong generalization. Performance metrics included loss, accuracy, precision, recall, and F1 score.

Although the overall accuracy was high, the confusion matrix showed a bias toward the Cucumber class, with frequent misclassifications of Cabbage, Carrot, and Tomato. This indicates a possible class imbalance or feature similarity.

ResNet-18 was chosen as the best model for this task due to its efficiency, residual learning structure, and strong performance with limited data.

Figure 2 shows the training accuracy. Figure 3 presents the validation accuracy, and Figure 4 illustrates the performance trend over the training epochs.

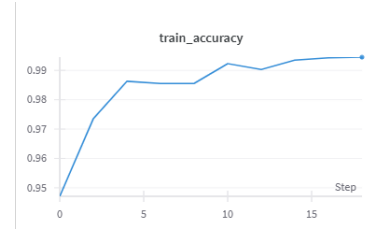


Figure 2: Training accuracy.

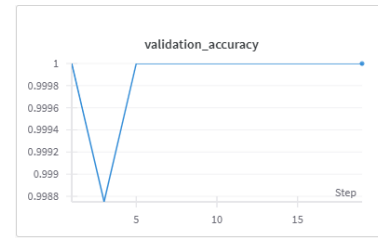


Figure 3: Validation accuracy.

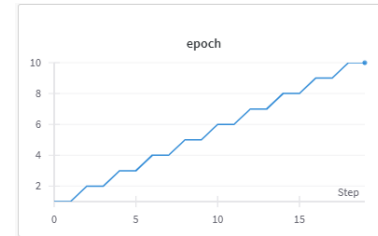


Figure 4: Performance trend over the 10 training epochs.

5 Discussion

The model performed well overall, with high accuracy and low loss, indicating good generalization. However, the confusion matrix revealed a skew toward the cucumber class, likely due to class imbalance or visual similarity. While ResNet-18 was effective, alternative architectures such as EfficientNet, DenseNet, and MobileNetV3 may enhance class-level accuracy. Hybrid models, along with techniques like data augmentation, can further improve distinction among similar classes and reduce performance disparity.

6 Conclusion

This study demonstrated the effectiveness of deep learning for multi-class vegetable classification, achieving high accuracy and robust performance across both training and validation

sets. Nevertheless, the confusion matrix revealed notable misclassifications, particularly a tendency to overpredict the cucumber class, suggesting challenges related to class imbalance or visual similarity.

ResNet-18 proved to be an appropriate model for the task; however, future research could explore alternative architectures such as EfficientNet, DenseNet, Vision Transformers, or MobileNetV3 to enhance class-level performance. Deployment using Gradio and Hugging Face Spaces facilitated real-time interaction and accessibility. Future work may focus on mitigating class bias through data augmentation or other strategies to achieve more balanced and reliable predictions across all classes.

References

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